

Predictive Maintenance for Industrial Equipment Using Machine Learning

1. Abstract

The objective of this project is to develop a machine learning model capable of predicting machine failures using sensor and operational data. Predictive maintenance helps industries reduce downtime, reduce maintenance costs, and improve operational efficiency.

2. Problem Statement

Unexpected machine failures can lead to production losses and increased maintenance costs. The goal of this project is to predict machine failures before they occur using machine learning techniques.

3. Dataset Description

Dataset Name: AI4I 2020 Predictive Maintenance Dataset

Number of Records: 10,000

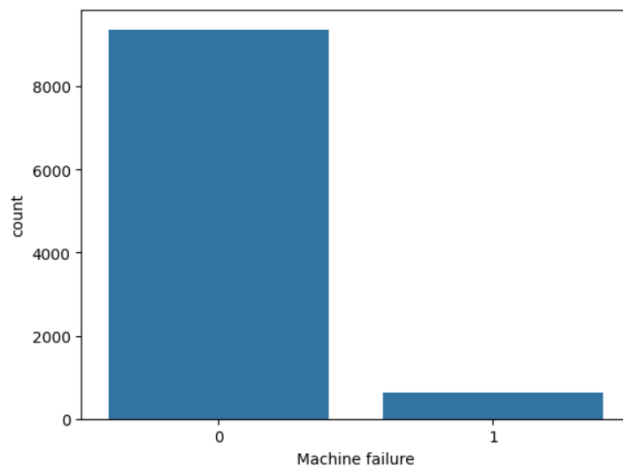
```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 8 columns):
 #   Column              Non-Null Count  Dtype  
---  -
 0   Air temperature [K]  10000 non-null  float64
 1   Process temperature [K]  10000 non-null  float64
 2   Rotational speed [rpm]  10000 non-null  int64  
 3   Torque [Nm]          10000 non-null  float64
 4   Tool wear [min]      10000 non-null  int64  
 5   Machine failure       10000 non-null  int64  
 6   Type_L               10000 non-null  bool    
 7   Type_M               10000 non-null  bool    
dtypes: bool(2), float64(3), int64(3)
memory usage: 488.4 KB
```

Target Variable:

- Machine Failure
 - 0 = No Failure
 - 1 = Failure

```
sns.countplot(x='Machine failure', data=df)
plt.show()
```



Features Used:

- Air Temperature [K]
- Process Temperature [K]
- Rotational Speed [rpm]
- Torque [Nm]
- Tool Wear [min]
- Machine Type

4. Data Preprocessing

The following preprocessing steps were performed:

- Checked for missing values
- Removed data leakage columns:
 - TWF
 - HDF
 - PWF
 - OSF
 - RNF
- Encoded the Type column using One-Hot Encoding
- Split the dataset into training and testing sets (80:20)

5. Exploratory Data Analysis (EDA)

The dataset was analyzed using descriptive statistics and visualization techniques.

Observations:

- Dataset contains 10,000 records.
- No missing values were found.
- Machine failures represent only 6.38% of the dataset, indicating class imbalance.

6. Models Implemented

Logistic Regression

Accuracy: 95%

Recall: 24%

```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred_lr))
```

	precision	recall	f1-score	support
0	0.95	1.00	0.97	1872
1	0.86	0.24	0.38	128
accuracy			0.95	2000
macro avg	0.91	0.62	0.68	2000
weighted avg	0.94	0.95	0.94	2000

Decision Tree

Accuracy: 95%

Recall: 66%

```
from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred_dt))
```

	precision	recall	f1-score	support
0	0.98	0.97	0.97	1872
1	0.60	0.66	0.62	128
accuracy			0.95	2000
macro avg	0.79	0.81	0.80	2000
weighted avg	0.95	0.95	0.95	2000

Random Forest

Accuracy: 97%

Precision: 95%

Recall: 54%

F1 Score: 69%

```
from sklearn.metrics import classification_report  
  
print(classification_report(y_test, y_pred_rf))
```

	precision	recall	f1-score	support
0	0.97	1.00	0.98	1872
1	0.95	0.54	0.69	128
accuracy			0.97	2000
macro avg	0.96	0.77	0.83	2000
weighted avg	0.97	0.97	0.96	2000

7. Cross Validation

Stratified K-Fold Cross Validation was used to handle class imbalance.

Mean F1 Score: 0.677

```
print(scores)  
print("Mean F1 Score:", scores.mean())
```

```
... [0.67692308 0.71568627 0.63917526 0.67346939 0.67980296]  
Mean F1 Score: 0.6770113905169932
```

8. Hyperparameter Tuning

Grid Search was applied to optimize the Random Forest model.

Best Parameters:

- n_estimators = 100
- max_depth = None

```
print(grid_search.best_params_)  
print(grid_search.best_score_)
```

```
{'max_depth': None, 'n_estimators': 100}  
0.6571570585216903
```

9. Confusion Matrix

Confusion Matrix:

```
[[1868, 4],  
 [59, 69]]
```

Interpretation:

- True Negatives: 1868
- False Positives: 4
- False Negatives: 59
- True Positives: 69

10. ROC-AUC Score

ROC-AUC Score: 0.94

The model demonstrates excellent discrimination capability between failure and non-failure classes.

ROC-AUC Score: 0.9401166866987181

11. Feature Importance

Top Factors Influencing Machine Failure:

1. Tool Wear [min]
2. Torque [Nm]
3. Rotational Speed [rpm]

feature_importance

...	Feature	Importance
4	Tool wear [min]	0.309174
3	Torque [Nm]	0.283466
2	Rotational speed [rpm]	0.140627
1	Process temperature [K]	0.119911
0	Air temperature [K]	0.119250
5	Type_L	0.017570
6	Type_M	0.010002

4.

12. Business Recommendations

- Monitor tool wear regularly.
- Track abnormal torque values.
- Control excessive rotational speeds.
- Schedule preventive maintenance.

- Use model predictions to reduce downtime and maintenance costs.

Prediction Demo

```
) sample = pd.DataFrame({
    'Air temperature [K]': [300],
    'Process temperature [K]': [310],
    'Rotational speed [rpm]': [1500],
    'Torque [Nm]': [40],
    'Tool wear [min]': [120],
    'Type_L': [1],
    'Type_M': [0]
})

prediction = rf.predict(sample)

if prediction[0] == 0:
    print("Prediction = 0")
    print("Machine Healthy")
else:
    print("Prediction = 1")
    print("Machine Failure Predicted")
```

- Prediction = 0
Machine Healthy

```
sample_failure = pd.DataFrame({
    'Air temperature [K]': [305],
    'Process temperature [K]': [315],
    'Rotational speed [rpm]': [2000],
    'Torque [Nm]': [70],
    'Tool wear [min]': [240],
    'Type_L': [1],
    'Type_M': [0]
})

prediction = rf.predict(sample_failure)

if prediction[0] == 0:
    print("Prediction = 0")
    print("Machine Healthy")
else:
    print("Prediction = 1")
    print("Machine Failure Predicted")
```

Prediction = 1
Machine Failure Predicted

13. Conclusion

A Random Forest model was developed for predictive maintenance using machine sensor data. The model achieved 97% accuracy and a ROC-AUC score of 0.94. Feature importance analysis revealed that Tool Wear, Torque, and Rotational Speed are the primary factors contributing to machine failures. The developed solution can help industries implement proactive maintenance strategies and reduce unexpected downtime.